

# Running Online User Studies with the reVISit Framework

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## Abstract

Running online user studies can be done using commercial survey tools or building custom software. However, these methods have limitations and require significant labor. This course introduces reVISit, an open-source alternative that streamlines study design. ReVISit eliminates tedious tasks by providing built-in components for UI, data hosting, participant recruiting, and more. Study designers can focus on research questions and stimulus design using a domain-specific language to quickly create studies as static websites. Throughout the course, participants will develop and deploy an interactive, fully instrumented study, improving their skills through hands-on experience with reVISit.

## CCS Concepts

• **Human-centered computing** → **User studies.**

## Keywords

User studies, Research methodology, Data collection and analysis

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## 1 Introduction

ReVISit [2] is an open platform designed to enable the visualization and HCI community to run sophisticated user studies through a web browser. ReVISit is a \$ 2 mio USD investment by the US National

Science Foundation into the visualization community. The proposed course is a critical component of the community engagement for this project: we seek to both educate the community about how to run online user studies with reVISit, and to recruit community members to use and provide feedback on reVISit. In its second year, reVISit has passed the prototype stage: it is well documented, comes with many examples, and has been used in classrooms and in numerous real studies, including several VIS/CHI submissions for this year's full and short papers tracks.

Through this course, we hope to show researchers what reVISit is, the problems that it solves, and how to use it. The course roughly follows the online tutorials that are available at <https://revisit.dev/>, with new content and activities for CHI attendees. Participants will first learn how to construct and deploy a basic study, then learn about data collection and storage, followed by an introduction to instrumenting interactive components. We will then introduce advanced topics, such as sophisticated randomization strategies (including attention checks, breaks, etc.), the integration with crowd-sourcing platforms, and best practices for reproducible science.

The course will mix lecture-style presentation, where concepts are introduced, with hands-on exercises based on prepared examples. At all times, we will have multiple team members assisting participants with issues they encounter.

## 2 Intended Audience and Benefits

Our ideal audience are researchers who want to run reproducible online user studies. Ideally, they would have some knowledge of JSON, web development, and study design principles, but our guided walkthroughs will assume no prerequisite knowledge. By the end of this course these participants will be able to use the reVISit framework to build complex, open, and reproducible studies with intricate, interactive stimuli and advanced randomization strategies. They will understand how to recruit participants to the platform and how to export their data for more in-depth analysis.

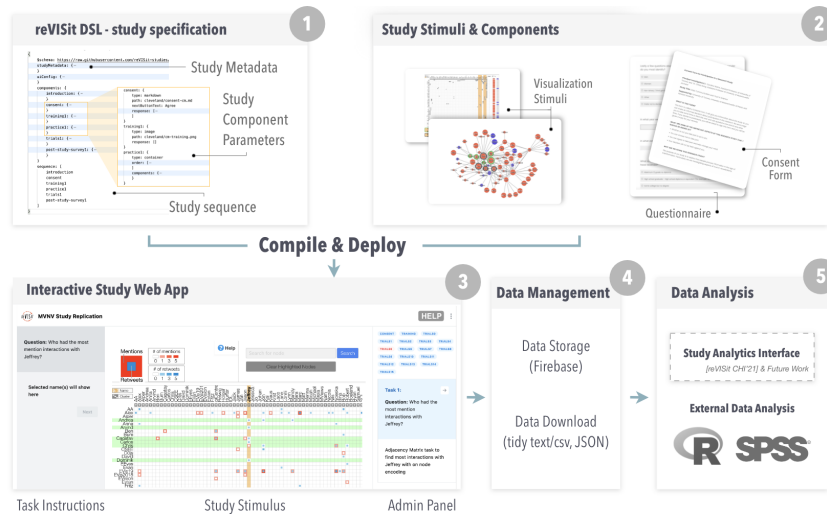
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**Figure 1: Overview of the revisit platform [2].** First, an experimenter generates a (1) study specification describing the study using the reVISit DSL. They also create (2) study stimuli and components such as consent forms and trainings. The specification and the study components are then compiled to an (3) interactive web app, which can be deployed to the web. The web app includes an admin interface for quickly browsing and debugging a study. Data from the study can either be (4) downloaded at the end of a trial, or stored on a server. Collected data can be (5) analyzed with external software such as R and SPSS, or examined and analyzed using the reVISit analytics interface.

### 3 Course Contents and Practical Work

*Introduction to reVISit (20 min).* We start the course with an introduction to reVISit. We first will address the challenges that currently exist for running visualization user studies in the browser, how the reVISit platform helps to alleviate these problems, and how it can be used to collect rich data. We will then discuss how reVISit provides data sovereignty to our users, and, finally, explore how reVISit supports open science.

*Getting Started with reVISit (30 min).* In this module, participants will learn how to clone, build, modify and deploy an existing study.

We will first take a deep dive into the reVISit.spec, a JSON-based Domain-Specific Language that reVISit uses to compile and build a study. Figure 1 shows an overview of the reVISit framework and the process for deploying a study.

The standard pipeline for working with reVISit is to start by forking and cloning the study repository. Once dependencies are installed, participants will see a list of demo studies which come with the reVISit repository.

We then continue by creating our first study components: an introduction and a help page. Once those are created, we have all the assets we need to create a simple questionnaire study. The user will then create a JSON configuration based on a template, and add survey questions. Then, the new study will be visible from the main dashboard of reVISit. It is important to note that ReVISit seeks to mitigate several of the pain points of experiment management and hosting. For this phase, participants are producing and editing simple text files, which reVISit compiles into a deployable study.

Based on this study, we will introduce course participants to reVISit’s admin functionalities, which target various pain points,

such as the ability to skip questions or trial blocks to rapidly debug experiment stimuli, to see the progress of the study, etc.

*Data Storage and Security (30 min).* One of our main goals for reVISit was to make sure that the data captured from study participants is both secure and is only managed by the study creator. No data is ever passed to the creators of reVISit which means that the study creator has complete sovereignty over the data that reVISit captures from participants.

When first cloning the repository, reVISit is set to use the local storage engine: study specific data is stored only in the browser and is available for download. This does not lend itself to deploying reVISit in a production environment. In order to create a platform which is as accessible as possible and still provides external storage for production environments, we use Firebase to provide a complete storage and security system that doesn’t require experimenters to run their own servers.

As a next step in the course, participants will set up a Firebase application and integrate this with their running reVISit example, simply by changing a few configurations in the repository. Participant data is then directly stored in Firebase. We will then introduce the admin dashboard where authorized users can download this user data without having to interact with Firebase directly.

ReVISit also uses Firebase’s Google SSO integration to enable authentication and authorization. We will demonstrate how to enable Google SSO in Firebase, so that only authorized users can access the study participant data.

*Creating Interactive Stimuli (40 min).* As a next step, participants will learn how to integrate interactive stimuli such as data visualizations into reVISit. We will first introduce the basic principles of how such stimuli are integrated with reVISit, and then break

out into two groups: one using vanilla HTML and JavaScript, the other one using React. Course participants can self-select for these groups based on their background knowledge and preferences. Both groups will be provided with scaffolds that they can modify and integrate in their ongoing example.

*Provenance Tracking for Interactive Stimuli (30 min).* reVISit automatically stores study-level provenance, such as start/end time of each task, responses, task order, and window events such as browsed in/out and mouse move. Such data are particularly useful in the study design process, for example to determine if participants are spending sufficient time in training and course content for a given study.

Additionally, reVISit allows for easy integration with Ttrack [1], a state-based provenance tracking library developed by our team. Ttrack can be used to store provenance on a task level, allowing researchers to make their own decisions of what parts of their system they want to track. If a study designer uses Ttrack in their stimuli, reVISit will show a timeline of all interactions with the stimuli from the participant. This provenance data stream enables teams to better understand any potential issues with their stimuli and how participants are completing tasks.

Being a state-based provenance library, Ttrack is easily integrated into React components. For more complex systems, a version of Ttrack designed as an extension to the popular React library, Redux, also exists. For vanilla HTML/Javascript users, Ttrack is integrated by defining a state your application wants to track, and updating both the state and the frontend via an observer pattern.

In the course, we will introduce Ttrack first with a simple example, and then again break out into two groups: participants who have chosen a React-based stimulus in the previous section will instrument this one, while others will instrument their vanilla HTML/JavaScript stimulus.

*Study Flow (30 min).* In this section, we will describe how reVISit manages study flow. Up to this point, we will have used only a linear sequence of study components, such as consent, training and stimuli. In this module, we will introduce more advanced strategies for study design, including randomization, breaks, and attention checks.

reVISit allows for multiple types of randomization and complex study design by allowing the user to create their study order via a system of blocks, which we will demonstrate throughout the course. Each block contains a list of trials, and has a defined order using a linear, random, or latin square approach. We will discuss when each randomization strategy is useful, and how to implement them. Blocks can be arbitrarily nested, allowing for any type of complex study design. Users can also define sampling for each block, enabling design such as ‘a participant sees 2 out of these 5 tasks’. Each block can also have interruptions added into the tasks, either randomly or at a consistent interval, which is useful for giving participants breaks, or for attention checks.

A common problem in a crowd-sourced study is handling complex randomization when a participant quits an ongoing study. We will demonstrate how reVISit overcomes such situations through, for example, automatically re-assigning incomplete latin square rows to later participants.

*Participant Recruiting (30 min).* Experiments built by reVISit support both in-lab experiments and online experiments. In this section, we will guide course participants through the process of setting up a crowdsourced experiment.

reVISit supports online recruiting through platforms like Amazon’s Mechanical Turk or Prolific [4]; but reVISit studies can also be distributed through email, social media, or platforms such as LabInTheWild [6]. In this section, we will guide attendees through an example on how to use Prolific to recruit participants for experiments built with reVISit, as it is both, a representative use case and also is increasingly used for visualization studies, given its sizable faceted participant pools and many features focused on empowering research [5].

*Data Extraction and Analysis (10 min).* The rich data that reVISit captures is only useful if a researcher can analyze it efficiently. To that end, the reVISit platform provides multiple ways to access and analyze the data. First, we will briefly discuss how reVISit allows users to download the raw data as both JSON and tidy CSV files. With this data, researchers can then use their own tool of choice to run their analyses, allowing arbitrarily complex statistical tests in a familiar analysis environment. Finally, we will give a brief introduction into the built-in analysis capabilities of the reVISit platform, which targets event sequence visualizations that leverage the captured provenance data [3].

*Best Practices for Open Science (10 min).* reVISit is designed to enable long-term reproducibility. In this course section, we will discuss how reVISit experiments can be anonymously shared with reviewers and the general public, so that everyone can evaluate exactly what happened in a study. When reVISit code and data is published, others can also easily re-use parts of experiments or re-analyze the data. We consider these aspects a key differentiator to commercial platforms like Qualtrics.

*reVISit Community and Roadmap (10 min).* reVISit is ultimately a community-focused open-source project. This course is part of a series of initiatives and planned events for the project. The reVISit team currently maintains Slack and GitHub organizations to support platform users and project development. In these forums, students and researchers have received support for edge-cases (e.g., unique study designs) and common infrastructure challenges (e.g., tool configuration). This effectively expands an individual research lab’s scope of expertise and support in conducting visualization studies. From this course we hope to provide a larger array of researchers with a solid foundation to conduct their own studies using this technology and to grow the reVISit community, benefiting the CHI community in the process.

Given reVISit’s open-source nature, we anticipate that community members will want to publish some of their modifications back upstream to the reVISit GitHub repository. Our team is ready to review pull requests submitted by our community, and ready to collaborate with our community members to build new features and fix bugs. Our development roadmap is publicly accessible from our GitHub organization.

## 4 Instructor Information

**Jack Wilburn** is a senior software engineer at the Scientific Computing and Imaging Institute and the School of Computing at the University of Utah.

**Hilson Shrestha** is a PhD student working with Lane Harrison at the VIEW Lab at the Worcester Polytechnic Institute.

**Zach Cutler** is a PhD student at the Visualization Design Lab at the University of Utah.

**Yiren Ding** is a PhD student at the VIEW Lab at Worcester Polytechnic Institute.

**Brian Bollen** is a senior software developer at the Scientific Computing and Imaging Institute at the University of Utah.

**Carolina Nobre** is an assistant professor of computer science at the University of Toronto. Her research focuses on creating user adaptive interactive data visualizations.

**Lane Harrison** is an associate professor of computer science in the Department of Computer Science at Worcester Polytechnic Institute. Lane builds visualization systems to better understand how people engage and use visualizations they encounter in day to day life. Lane has built prior visualization experiment frameworks (experimentr) which has been used in 10+ VIS and CHI papers

beyond his own work and served as part of the inspiration for the reVISit initiative.

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